

WM — What's in a Medicine?

OCR B (Salters) H433 storyline **WM**. From willow bark to aspirin: the chemistry of the -OH group in alcohols and phenols, carboxylic acids and esters, IR and mass spectrometry, preparative techniques, chromatography and green chemistry.

1. Alcohols

Classification

Primary alcohol (1°): The -OH group is attached to a carbon bonded to one other carbon (or none, as in methanol).

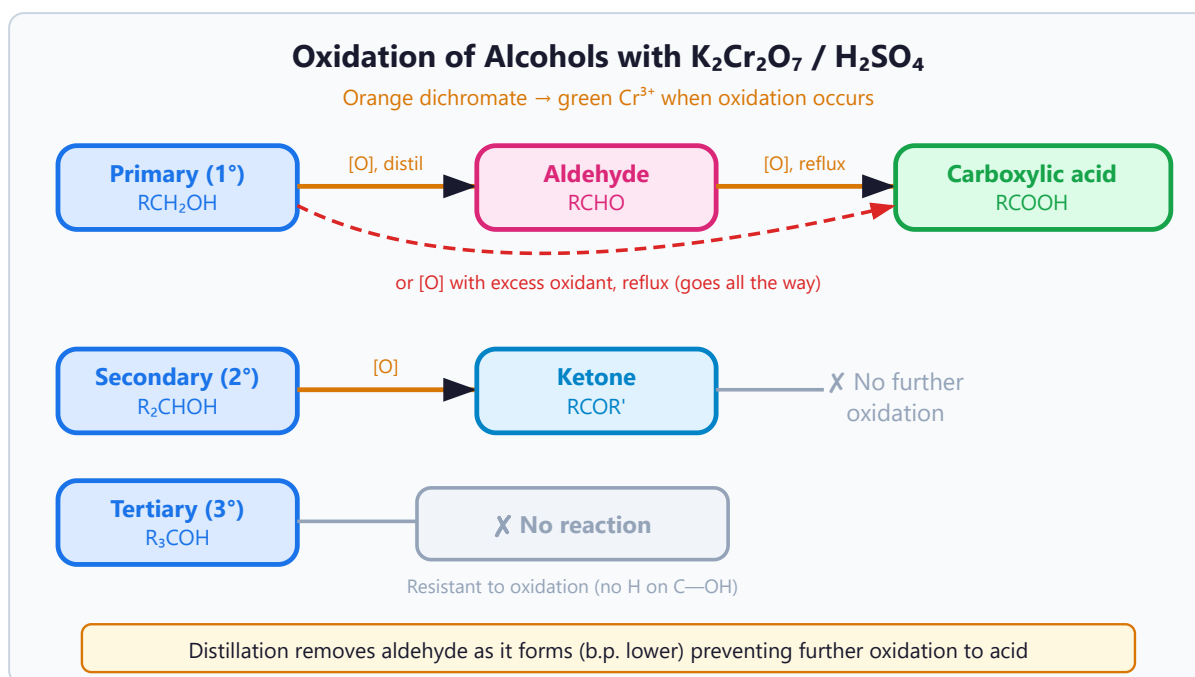
Secondary alcohol (2°): The -OH group is attached to a carbon bonded to two other carbons.

Tertiary alcohol (3°): The -OH group is attached to a carbon bonded to three other carbons.

Oxidation of Alcohols

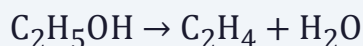
The oxidising agent is acidified potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$). The dichromate changes from orange to green ($\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+}) when oxidation occurs. The oxidising agent is represented as [O] in equations.

Alcohol type	Product with distillation	Product with reflux
Primary (1°)	Aldehyde (distil immediately to prevent further oxidation)	Carboxylic acid
Secondary (2°)	Ketone (cannot be further oxidised)	
Tertiary (3°)	No reaction (resistant to oxidation under these conditions)	



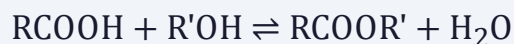
Dehydration

Alcohols can be dehydrated to form alkenes by heating with a strong acid catalyst (concentrated H_2SO_4 or H_3PO_4) or by passing alcohol vapour over heated aluminium oxide (Al_2O_3). Water is eliminated:



Esterification

Alcohols react with carboxylic acids in the presence of a concentrated sulfuric acid catalyst to form esters and water (a condensation reaction):



This is a reversible reaction. Esters have characteristic fruity smells and are used as flavourings, solvents, and in making plasticisers.

2. Phenols

Willow bark contains salicin, and the compound extracted from it in the nineteenth century — salicylic acid — carries two $-OH$ groups of different kinds. One is a carboxylic acid; the other is attached straight onto

the benzene ring. That second kind behaves quite unlike the alcohols of section 1, and the difference is the reason aspirin has to be made the way it is.

Phenol: A compound in which an -OH group is bonded *directly* to a carbon atom of a benzene ring. The simplest member is phenol itself, $\text{C}_6\text{H}_5\text{OH}$.

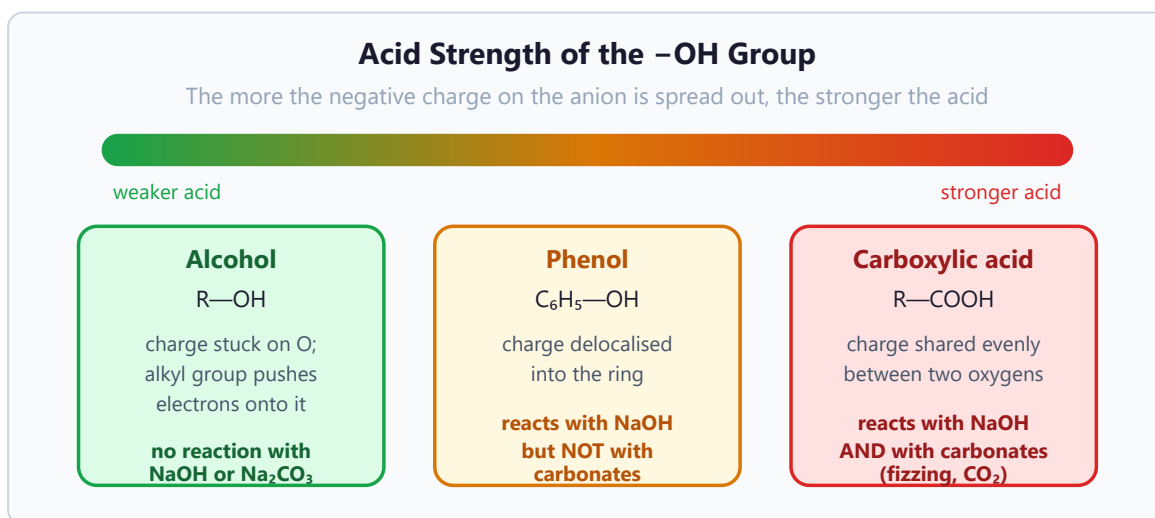
Be careful with the distinction. In phenylmethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$, the -OH sits on a CH_2 group next to the ring, not on the ring itself — that compound is an ordinary alcohol and shows none of the reactions below. Only an -OH joined straight to the ring carbon counts.

Why Phenols Are Acidic

Alcohols are essentially neutral. Phenols are genuinely, if weakly, acidic: a solution of phenol in water has a pH of about 5 to 6 and turns universal indicator orange.

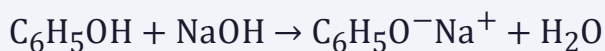


The explanation is the stability of the ion that is left behind. In the phenoxide ion, a lone pair on the oxygen overlaps with the delocalised π system of the ring, so the negative charge is spread out over the oxygen *and* the ring rather than being concentrated on one atom. A spread-out charge is a more stable charge, so the equilibrium above lies further to the right than it would for an alcohol, where the alkoxide ion RO^- has nowhere to put the charge and is further destabilised by the electron-donating alkyl group.



Reaction with Alkalis but Not with Carbonates

Phenol is acidic enough to be neutralised by a strong base. Add sodium hydroxide solution to the cloudy suspension you get when phenol is shaken with water and it clears, because the ionic salt sodium phenoxide is fully soluble:



But phenol is *not* acidic enough to displace carbon dioxide from a carbonate. Add sodium carbonate solution and nothing happens: no fizzing, no gas. A carboxylic acid, being the stronger acid of the two, does react:



This single difference is the standard way of telling a phenol from a carboxylic acid. Both dissolve in sodium hydroxide; only the carboxylic acid effervesces with sodium carbonate or sodium hydrogencarbonate.

The Neutral Iron(III) Chloride Test

The positive, colour-producing test for a phenol is to add a few drops of **neutral iron(III) chloride solution**. A phenol gives an immediate intense **purple** (violet) colouration; alcohols and carboxylic acids do not.

The solution must be neutral, because iron(III) chloride solution is acidic as supplied and the colour only develops in the absence of excess acid — in practice a little sodium carbonate is added to the reagent first. The purple comes from a complex formed between Fe^{3+} and the phenoxide oxygen.

Telling the Three Groups Apart

Test	Alcohol, R-OH	Phenol, Ar-OH	Carboxylic acid, R-COOH
pH of aqueous solution	Neutral (7)	Weakly acidic (about 5–6)	Acidic (about 3–4)
Add NaOH(aq)	No reaction	Reacts — dissolves to form the phenoxide salt	Reacts — forms the carboxylate salt

Add Na ₂ CO ₃ (aq)	No reaction	No reaction	Effervescence; CO ₂ turns limewater milky
Add neutral FeCl ₃ (aq)	No purple colour	Purple colouration	No purple colour
Warm with carboxylic acid + conc. H ₂ SO ₄	Ester formed	No ester formed	—
Add an acid anhydride	Ester formed	Ester formed	—

Esterification with Acid Anhydrides

A phenol is a much poorer nucleophile than an alcohol, for exactly the reason that makes it a better acid: the oxygen lone pair is drawn into the ring and so is less available to attack a carbonyl carbon. In consequence a phenol will *not* form an ester with a carboxylic acid, however long you reflux it with concentrated sulfuric acid. It needs a more reactive acylating agent, and the one used industrially is an **acid anhydride**.

An acid anhydride has the general structure RCO–O–COR, formally two carboxylic acid molecules minus one water. The commonest is ethanoic anhydride, (CH₃CO)₂O. The ester forms and a molecule of carboxylic acid is released:



The product, phenyl ethanoate, is the ester of phenol and ethanoic acid — made without ever letting the two meet.

Worked Example — The Synthesis of Aspirin:

Salicylic acid (2-hydroxybenzoic acid) has a phenol –OH on the ring and a –COOH group next to it. It is the phenol group that is esterified:



Check the balance: C₇H₆O₃ + C₄H₆O₃ gives C₁₁H₁₂O₆ on the left; C₉H₈O₄ + C₂H₄O₂ gives C₁₁H₁₂O₆

on the right. ✓

Because ethanoic anhydride is used rather than ethanoic acid, the reaction goes essentially to completion instead of settling at an equilibrium — and the -COOH group of the salicylic acid, which is not a phenol, is left untouched.

There is a neat way to check that the reaction has worked. Salicylic acid contains a free phenol group, so it gives a purple colour with neutral iron(III) chloride. In aspirin that phenol group has been converted into an ester, so pure aspirin gives *no* purple colour. A purple tinge in the product means unreacted salicylic acid is still present and the crystals need recrystallising again.

Exam Tip: Two answers are worth rehearsing word for word. "Why is phenol a stronger acid than ethanol?" — because the negative charge of the phenoxide ion is delocalised over the benzene ring, making the ion more stable, so the equilibrium releasing H^+ lies further to the right. "How would you distinguish phenol from benzoic acid?" — add sodium carbonate solution; the carboxylic acid effervesces and the phenol does not. Do not offer sodium hydroxide for that one, since both react with it, and do not offer sodium metal, since both release hydrogen.

3. Carboxylic Acids and Derivatives

Carboxylic Acids

Carboxylic acids contain the -COOH functional group. They are weak acids that partially dissociate in water: $\text{RCOOH} \rightleftharpoons \text{RCOO}^- + \text{H}^+$. They react with bases, carbonates, and reactive metals to form salts (carboxylates).

Esters

Esters (RCOOR') are formed by the condensation reaction between a carboxylic acid and an alcohol (Fischer esterification) in the presence of an acid catalyst (concentrated H_2SO_4). This is a reversible reaction.

Hydrolysis of esters:

- **Acid hydrolysis:** Refluxing with dilute acid (e.g. dilute H_2SO_4) breaks the ester back into the carboxylic acid and alcohol. This is reversible.

- **Base hydrolysis (saponification):** Refluxing with dilute NaOH produces the sodium salt of the carboxylic acid (carboxylate) and the alcohol. This is irreversible because the carboxylate ion is very stable.

Acyl Chlorides

Acyl chloride (RCOCl): A reactive derivative of carboxylic acids in which the OH of the COOH group is replaced by Cl. They are much more reactive than carboxylic acids because the C–Cl bond is weaker and the Cl is a good leaving group.

Acyl chlorides react vigorously at room temperature with:

Reagent	Product	Other product
Water (H ₂ O)	Carboxylic acid	HCl
Alcohol (ROH)	Ester	HCl
Ammonia (NH ₃)	Amide (RCONH ₂)	HCl
Amine (R'NH ₂)	N-substituted amide (RCONHR')	HCl

These are all nucleophilic addition-elimination reactions. The nucleophile (H₂O, ROH, NH₃, or R'NH₂) attacks the δ^+ carbonyl carbon, and the Cl[−] leaves.

4. Organic Synthesis and Retrosynthesis

Organic synthesis: Planning a series of reactions to convert a starting material into a desired target molecule.

Retrosynthesis: Working backwards from the target molecule to identify simpler starting materials and the reactions needed. The target is disconnected at key bonds to identify precursor molecules.

Key synthetic routes to know (building a "reaction map"):

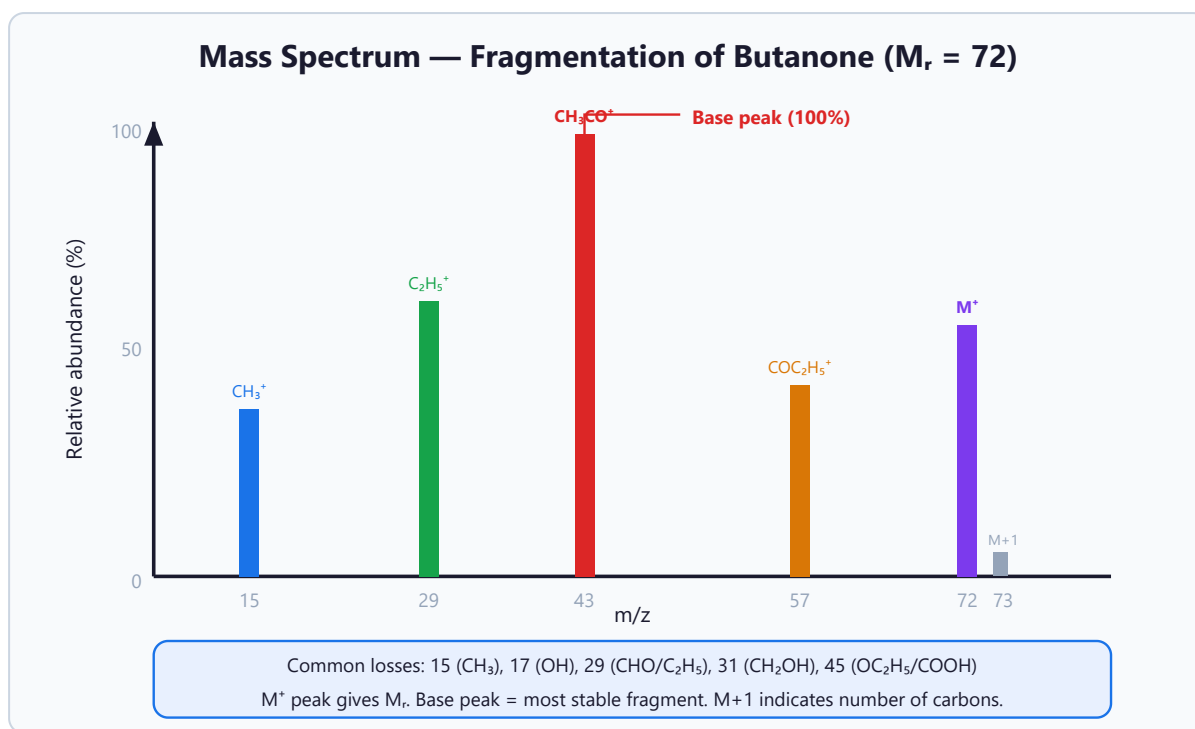
- Alkene → alcohol (hydration with H_3PO_4 catalyst, or addition of H_2O via H_2SO_4)
- Alcohol → halogenoalkane (HBr , or $\text{NaBr} + \text{H}_2\text{SO}_4$, or $\text{PCl}_3/\text{PCl}_5/\text{SOCl}_2$)
- Halogenoalkane → amine (excess NH_3 , sealed tube)
- Halogenoalkane → nitrile (KCN in ethanol/water) → carboxylic acid (acid hydrolysis) or amine (LiAlH_4 reduction)
- Alcohol → aldehyde (oxidation, distil) → carboxylic acid (further oxidation, reflux)
- Carboxylic acid + alcohol → ester (acid catalyst)
- Acyl chloride + amine → amide
- Benzene → nitrobenzene (nitration) → phenylamine (reduction with Sn/HCl)

5. Analytical Techniques

Mass Spectrometry

Mass spectrometry identifies compounds by their molecular mass and fragmentation pattern. The molecular ion peak (M^+) gives the relative molecular mass. Fragmentation peaks correspond to pieces of the molecule after bonds break. Common fragment losses: 15 (CH_3), 17 (OH), 29 (CHO or C_2H_5), 31 (CH_2OH), 45 (OC_2H_5 or COOH).

The $\text{M}+1$ peak (one mass unit above M^+) is caused by the presence of ^{13}C . The relative height of the $\text{M}+1$ peak indicates the number of carbon atoms in the molecule.

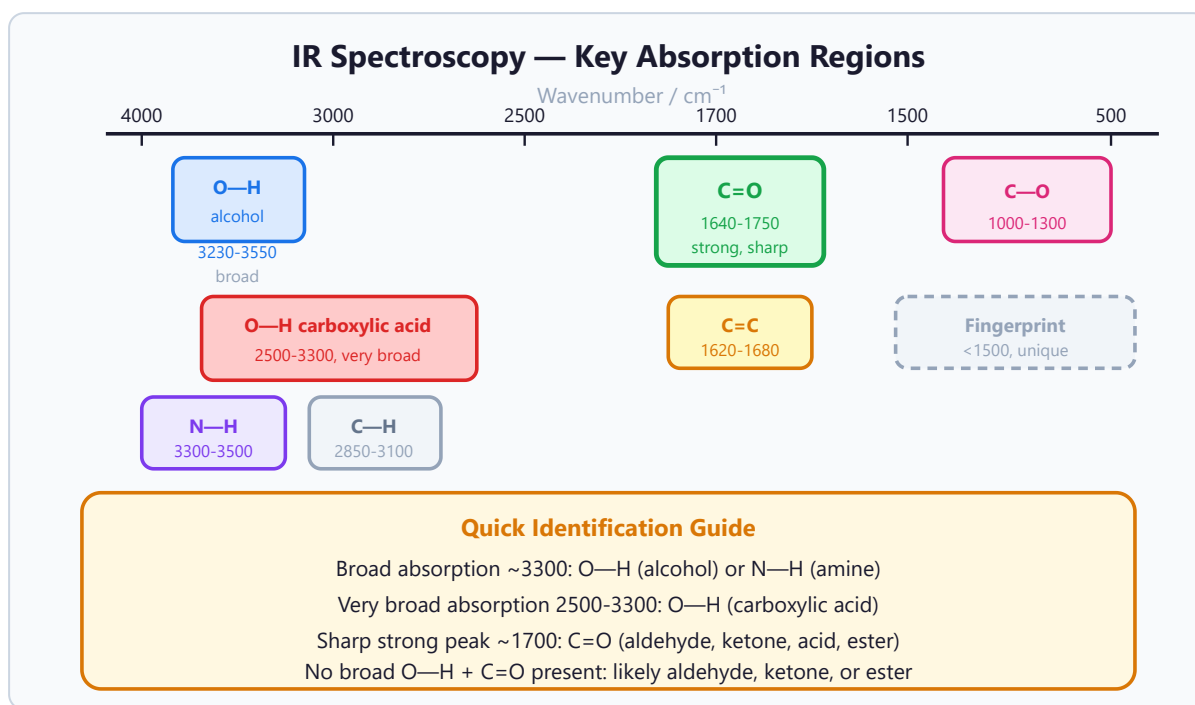


Infrared (IR) Spectroscopy

IR spectroscopy: Identifies functional groups by the absorption of infrared radiation, which causes covalent bonds to vibrate (stretch and bend) at characteristic frequencies. An IR spectrum plots transmittance (%) against wavenumber (cm^{-1}).

Bond	Wavenumber (cm^{-1})	Appearance
O–H (alcohol)	3230–3550	Broad
O–H (carboxylic acid)	2500–3300	Very broad
N–H (amine)	3300–3500	Medium, may show 2 peaks
C–H	2850–3100	Medium to strong
C=O	1640–1750	Strong, sharp
C=C	1620–1680	Medium
C–O	1000–1300	Strong

The fingerprint region (below about 1500 cm^{-1}) is unique to each compound and can be compared against a database for identification. IR spectroscopy can also be used to monitor the progress of a reaction by observing the disappearance or appearance of characteristic absorptions.



NMR Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy: Detects the chemical environments of specific nuclei (usually ^1H or ^{13}C) in a magnetic field. Nuclei in different chemical environments absorb at different frequencies, giving peaks at different chemical shifts (δ , measured in ppm relative to TMS).

Tetramethylsilane (TMS, $\text{Si}(\text{CH}_3)_4$) is the reference standard ($\delta = 0\text{ ppm}$) because it is chemically inert, gives a single sharp peak (all 12 H atoms are equivalent), and its peak is well away from most other signals.

^{13}C NMR

^{13}C NMR shows one peak for each unique carbon environment in the molecule. The number of peaks tells you how many different carbon environments there are. Symmetry can reduce the number of peaks (equivalent carbons give only one peak). Chemical shift values indicate the type of carbon environment:

Carbon environment	Approximate δ / ppm
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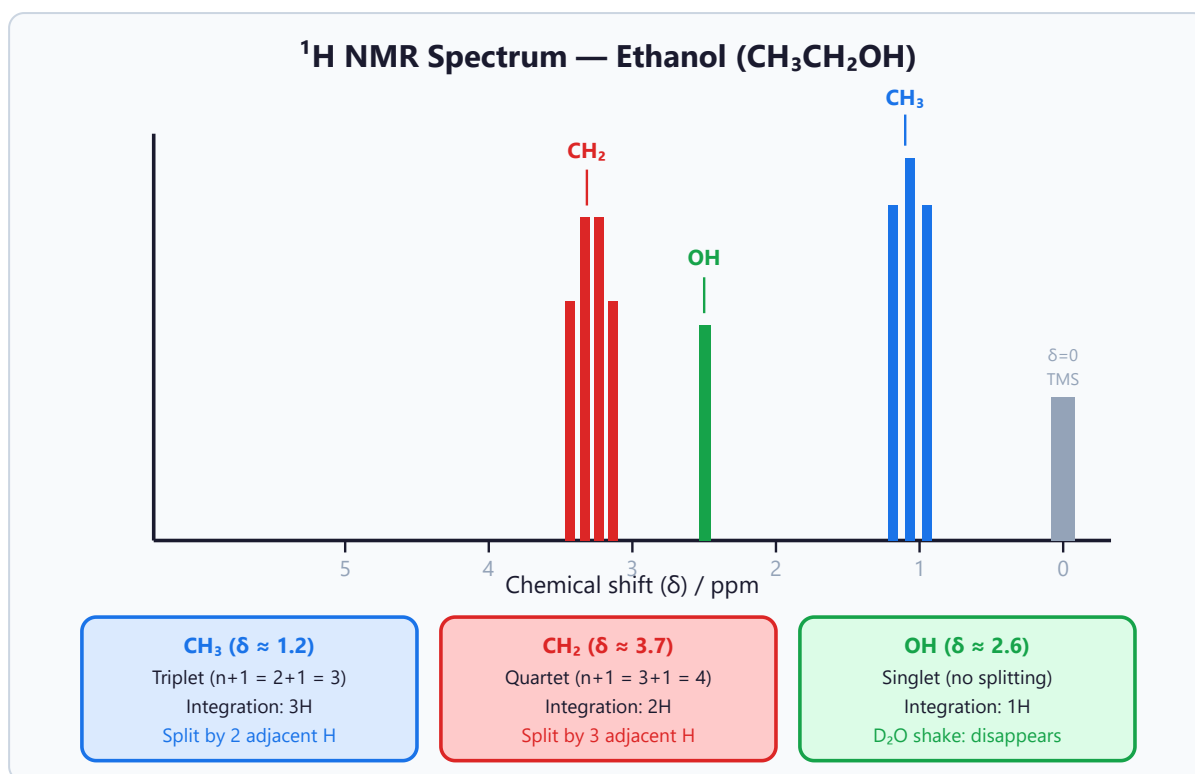
R-CH ₃ , R-CH ₂ -R (alkyl)	5–40
C-Cl, C-Br	25–65
C-O (alcohols, esters)	50–80
C=C (aromatic)	110–160
C=O (carbonyl)	170–220

¹H NMR

¹H NMR is richer in information than ¹³C NMR because it provides three pieces of data for each signal:

1. **Chemical shift (δ):** Indicates the chemical environment of the hydrogen atoms.
2. **Integration (relative peak area):** Proportional to the number of hydrogen atoms in that environment.
3. **Splitting pattern (multiplicity):** Determined by the number of hydrogen atoms on adjacent carbon atoms (the n+1 rule).

Proton environment	Approximate δ / ppm
R-CH ₃	0.7–1.2
R-CH ₂ -R	1.2–1.4
R-CH ₂ -X (halogen or O)	3.1–4.2
R-O-H (alcohol)	1.0–5.5 (variable)
Ar-H (aromatic)	6.5–8.0
R-CHO (aldehyde)	9.4–10.0
R-COOH (carboxylic acid)	10.0–12.0



Splitting (Spin-Spin Coupling)

The **$n+1$ rule**: A proton signal is split into $(n+1)$ peaks, where n is the number of equivalent protons on adjacent carbon atoms.

Adjacent equivalent H atoms (n)	Splitting pattern	Name
0	1 peak	Singlet (s)
1	2 peaks	Doublet (d)
2	3 peaks	Triplet (t)
3	4 peaks	Quartet (q)

The relative intensities follow Pascal's triangle: singlet 1, doublet 1:1, triplet 1:2:1, quartet 1:3:3:1. O–H and N–H protons usually appear as singlets (no splitting) because rapid exchange with the solvent averages out coupling. A D₂O shake (adding deuterium oxide) causes O–H and N–H peaks to disappear, confirming their identity.

Exam Tip: When interpreting ^1H NMR spectra: (1) count the number of signals to determine how many different H environments exist; (2) use integration ratios to find the relative number of H atoms; (3) use chemical shifts to identify the environments; (4) use splitting patterns to determine connectivity. Work systematically through all the data before proposing a structure.

6. Chromatography

Thin-Layer Chromatography (TLC)

TLC uses a thin layer of silica gel or alumina on a plate (the stationary phase). The sample is spotted near the bottom, and a solvent (mobile phase) rises up the plate by capillary action. Components separate based on their relative affinities for the stationary and mobile phases.

$$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$$

R_f values are characteristic of a substance under specific conditions and can be compared with standards for identification. TLC is quick and useful for monitoring reaction progress and checking purity.

Gas Chromatography (GC)

GC separates volatile compounds. The sample is vaporised and carried through a long, coiled column by an inert carrier gas (the mobile phase). The column is packed with a high-boiling-point liquid on an inert solid support (the stationary phase). Components separate based on their boiling points and interactions with the stationary phase. A detector records the signal as each component elutes, producing a chromatogram where peak retention time identifies the component and peak area is proportional to amount.

High-Performance Liquid Chromatography (HPLC)

HPLC is used for non-volatile or thermally unstable compounds that cannot be analysed by GC. A liquid mobile phase is pumped at high pressure through a column packed with fine particles (stationary phase). The technique gives excellent separation and is widely used in pharmaceutical analysis. A UV detector is commonly used. Like GC, retention time identifies components and peak area gives quantitative information.

GC-MS and HPLC-MS combine chromatographic separation with mass spectrometry for both separation and identification in a single analysis, making them extremely powerful analytical tools.

Exam Tip: You may be asked to combine data from mass spectrometry, IR, and NMR to identify an unknown compound. Approach this systematically: (1) use the mass spectrum to find the molecular mass and key fragments; (2) calculate the degree of unsaturation ($\text{IHD} = (2C + 2 + N - H - X) / 2$; IHD of 4 strongly suggests a benzene ring); (3) use IR to identify functional groups; (4) use ^{13}C NMR to count carbon environments; (5) use ^1H NMR to determine hydrogen environments, numbers, and connectivity. Propose a structure consistent with all the data.

Worked Example — Structure Determination:

Unknown compound: $M^+ = 72$. Molecular formula $\text{C}_4\text{H}_8\text{O}$. $\text{IHD} = (8 + 2 - 8) / 2 = 1$ (one $\text{C}=\text{O}$ or one ring).

IR: strong sharp absorption at 1720 cm^{-1} ($\text{C}=\text{O}$), no broad $\text{O}-\text{H}$ → ketone or aldehyde, not acid or alcohol.

^1H NMR: three peaks. δ 1.0 (triplet, 3H), δ 2.1 (singlet, 3H) and δ 2.4 (quartet, 2H). No peak at δ 9–10, so not an aldehyde.

The 3H singlet at δ 2.1 is a CH_3 next to $\text{C}=\text{O}$ with no adjacent H atoms; the 2H quartet and 3H triplet show a CH_2CH_3 group, with the CH_2 next to $\text{C}=\text{O}$.

Structure: butanone ($\text{CH}_3\text{COCH}_2\text{CH}_3$). Always cross-check all data against your proposed structure.

7. Green Chemistry

Green chemistry: Designing chemical products and processes so as to reduce or eliminate the use and generation of hazardous substances — and, more broadly, to minimise waste, energy use and damage to the environment across the whole life of the product.

Green chemistry is not a separate branch of the subject but a set of questions to ask about any process: what goes in, what comes out, how much energy it takes, and what happens to the product at the end of its life. The aspirin synthesis that closes this storyline is the case study H433 uses.

The Principles, Grouped

Twelve principles of green chemistry are widely quoted. You are **not** expected to recite them — the specification asks you to make suggestions based on them and to analyse information you are given — but they group naturally into five themes:

Theme	What it asks	In practice
Waste	Prevent waste rather than clean it up afterwards; design for a high atom economy; avoid unnecessary derivatives such as protecting groups	Choose an addition reaction over a substitution where possible; find a use for the by-product rather than disposing of it
Hazard	Use and generate substances of low toxicity; design syntheses that cannot run away; avoid hazardous solvents and auxiliaries	Replace benzene and chlorinated solvents with water, ethanol or supercritical CO ₂
Energy	Run reactions at ambient temperature and pressure where you can	Use a catalyst so the reaction goes at a lower temperature; recover waste heat
Feedstocks and catalysis	Use renewable raw materials; prefer catalytic reagents to stoichiometric ones	A catalyst is used over and over; a stoichiometric reagent is consumed and becomes waste. Enzymes work at body temperature and are highly selective
Life cycle and control	Design products that break down harmlessly; monitor processes in real time to prevent the formation of by-products	Biodegradable polyesters instead of persistent addition polymers; in-line analysis to stop a batch going wrong

Atom Economy

Percentage yield tells you how much of the theoretical product you actually isolated. It says nothing about how much of the starting material was *designed* to end up as waste. Atom economy does.

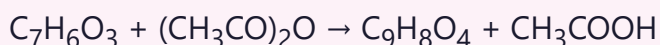
$$\% \text{ atom economy} = (\text{M}_r \text{ of the desired product} / \text{sum of M}_r \text{ of all products}) \times 100$$

A reaction can have a 100% yield and a poor atom economy, or a good atom economy and a poor yield. Both numbers matter: a high atom economy means less raw material is wasted and less waste has to be treated, which saves money as well as environmental damage. Addition reactions have an atom economy of 100% by definition, because there is only one product; substitution and elimination reactions never do.

Worked Example — Two Routes to Aspirin:

Aspirin ($\text{C}_9\text{H}_8\text{O}_4$, $M_r = 180$) is made by acylating salicylic acid (2-hydroxybenzoic acid). Compare the atom economy of the two possible acylating agents.

Route 1 — ethanoic anhydride:



Products: aspirin (180) and ethanoic acid (60)

$$\text{Atom economy} = 180 / (180 + 60) \times 100 = \mathbf{75.0\%}$$

Route 2 — ethanoyl chloride:



Products: aspirin (180) and hydrogen chloride (36.5)

$$\text{Atom economy} = 180 / (180 + 36.5) \times 100 = \mathbf{83.1\%}$$

Route 2 has the better atom economy — and yet industry uses Route 1. Atom economy is only one criterion.

Why Aspirin is Made with the Anhydride

That worked example is the point of the case study: a green-chemistry judgement is a balance, not a single number.

	Ethanoic anhydride	Ethanoyl chloride
Atom economy	75.0%	83.1%
By-product	Ethanoic acid — a liquid that can be recovered and used to make more anhydride	Hydrogen chloride — a corrosive, choking gas that must be scrubbed

Reaction with water	Reacts slowly; safe to handle with normal precautions	Reacts violently, giving off HCl fumes; hard to handle safely on a large scale
Cost and availability	Cheaper and made on a very large scale	More expensive

So the anhydride wins on hazard, on cost and on what happens to the by-product, and loses only on atom economy. The greener process is the one that is better *overall*.

A Larger Example: Ibuprofen

Ibuprofen was originally manufactured by Boots in a six-step synthesis with an atom economy of about 40% — three-fifths of the mass of the reactants ended up as waste. The BHC process, developed in the 1990s and recognised by a US Presidential Green Chemistry Challenge award in 1997, makes the same molecule in three catalytic steps with an atom economy of about 77%, rising higher still when the ethanoic acid by-product is recovered and reused. Fewer steps also means fewer solvents, fewer purifications and less energy.

The general lesson is worth carrying into any question: the biggest green gains usually come from *redesigning the route*, not from squeezing a few more percent of yield out of the route you already have.

Applying It in the Laboratory

The same thinking applies to a school preparation:

- Heating under reflux with a condenser rather than in an open flask stops volatile material escaping — less waste and less exposure.
- Recrystallising from the minimum volume of hot solvent, and recovering that solvent by distillation, reduces solvent use.
- Filtration under reduced pressure recovers more of the solid product, raising the yield.
- Reduced-scale apparatus uses less of everything and generates less waste, with no loss of chemistry.
- Choosing the catalysed route lets the reaction run at a lower temperature, saving energy.

Exam Tip: Green chemistry questions on this paper almost always supply the information — two routes, some M_r values, a note on hazards — and ask you to evaluate. Do not try to recall the twelve principles. Instead, calculate the atom economy where the data allows, then comment on at least two

other factors (hazard of the reagents, what the by-product is and whether it can be reused, energy demand, whether the feedstock is renewable, whether a catalyst is used). A conclusion that weighs the factors against each other will score far better than a list.